

The Macro-Evolutionary Theory of Outsourced Homeostasis

Why the brain seeks plant chemistry, and what that means for treating addiction

A framework draft · Andrew Morton

Biology has a habit of turning permanent environmental chemicals into functional necessities. What starts as a toxin can, over enough generations, become a tool — and then a requirement. This document argues that human drug-seeking behavior is the same pattern playing out with plant and fungal chemistry, and that what modern medicine calls “addiction” is better understood as a foraging drive operating in an environment it was never designed for.

1. The Great Oxidation Blueprint

The clearest precedent comes from 2.4 billion years ago. Cyanobacteria began producing oxygen as a metabolic waste product. To the anaerobic life that dominated the planet, O_2 was a corrosive poison — reactive enough to cause one of the largest extinction events in the history of life.

What happened next set a template biology has reused ever since:

- **Tolerance.** Surviving organisms evolved cellular defenses — catalase, superoxide dismutase, and related enzymes — just to live alongside the toxin without dying.
- **Exploitation.** Then came the leap. Complex life evolved aerobic respiration, using oxygen’s reactivity as an electron acceptor. The same property that made it poisonous made it the most powerful energy currency available — roughly fifteen times more ATP per glucose molecule than anaerobic pathways could extract.

Today, every cell in our body is completely dependent on what was once a planet-wide toxin. This is the macro-evolutionary filter: *tolerate* → *exploit* → *require*.

2. Deep-Time Coevolution with Plant Chemistry

The same filter ran a faster pass over mammals and the secondary metabolites of plants and fungi.

Land plants evolved alkaloids — caffeine, nicotine, morphine, and thousands of others — and fungi evolved their own arsenal. These were not made for us. They evolved as defenses against insects, designed to paralyze, deter, or kill predators at the cellular level.

But mammalian ancestors lived inside these chemical environments for tens of millions of years, and the same three-phase filter operated:

- **Tolerance.** Mammals evolved an unusually elaborate Cytochrome P450 enzyme system — a family of liver enzymes whose primary job is breaking down plant and fungal toxins. This system is large, redundant, and varies significantly between human populations.
- **Outsourcing.** Over deep time, the brain found it metabolically cheaper to let ambient environmental molecules tune certain signaling systems rather than manufacture every regulatory chemical internally. Ancient receptor families — the CB1 cannabinoid receptor (roughly 500 million years old), the opioid receptors, nicotinic acetylcholine receptors — became partial dependents on plant mimics for homeostatic stability.
- **The mineral lick analogy.** Mountain goats walk miles to lick specific rocks for sodium and magnesium. The behavior looks compulsive because it is — but it's correcting a real biological deficit. Hunter-gatherer humans were doing the same thing across a far wider chemical palette, ingesting micro-quantities of hundreds of compounds through diverse daily foraging.

3. The Mismatch

For roughly 95% of human history, humans lived plugged into this open, diverse chemical matrix. The brain's mesolimbic dopamine system — what neuroscientists call the SEEKING circuit — evolved to reward the discovery of these compounds because, at background micro-doses, they conferred real survival advantages:

- Stimulants suppressed fatigue and conserved caloric energy.
- Sedatives and anxiolytics buffered systemic inflammation and damped cortisol.
- Bitter antiparasitic compounds cleared internal worms and other parasites.

Then two things changed in rapid succession. Agriculture, about 10,000 years ago, narrowed the dietary chemical palette dramatically. Modern chemistry, in the last 200 years, learned to extract single molecules at 99%+ purity.

Ancestral baseline → Diluted micro-quantities, hundreds of compounds, wild forage.

Modern environment → Isolated macro-doses, single molecules, refined chemistry.

The addict's brain is doing exactly what natural selection designed it to do: scanning the environment for a compounding chemical resource and responding when it finds one. The problem is not the brain. The problem is that modern chemistry has hyper-concentrated the signal beyond anything the feedback loop was ever calibrated for. The brain reads a refined opioid the way a mineral-deficient goat would

read a salt mine the size of a stadium — as an unprecedented evolutionary jackpot, worth pursuing at any cost.

4. Heritage Nutrigenomics — The Clinical Implication

If the framework is right, the conclusion follows directly: modern chemical hunger is a deficiency state caused by rapid disconnection from an ancestral chemical baseline.

This is not a uniform deficiency. Human populations adapted to wildly different continental ecosystems over hundreds of thousands of years, and Cytochrome P450 variants and receptor populations vary accordingly. A lineage that evolved in a region dense with a particular botanical chemistry will have a genome that structurally expects its background presence — and feels its absence.

The therapeutic implication is a different shape than current addiction medicine:

1. **Catalog the ancestral chemistry.** Map continental-scale human genomes to the plant and fungal compounds they coevolved with.
2. **Reintroduce at micro-dose.** Supplement the modern diet with safe, regulated, heritage-specific quantities of those missing molecules — delivered in standard food forms rather than as isolated pharmaceuticals.
3. **Quiet the seeking drive.** Continuously satisfying the brain's homeostatic expectation resolves the deficit. The frantic foraging signal that drives compulsive macro-dose use stabilizes on its own, because the system isn't starving anymore.

This is preventative rather than punitive. It treats the addicted brain as a correctly functioning forager in a broken chemical environment, and proposes fixing the environment rather than punishing the forager.

Supporting literature

Evolutionary models of human drug use — overview of the ‘paradox of drug reward’ and the neurobiological substrates that respond to plant secondary metabolites. en.wikipedia.org/wiki/Evolutionary_models_of_human_drug_use

Sullivan, R. J., & Hagen, E. H. — work on psychotropic substance-seeking as evolved behavior, including the mineral-lick framing and antiparasitic hypotheses. pmc.ncbi.nlm.nih.gov/articles/PMC2367444/

Hagen et al. — nicotine and the antiparasitic hypothesis in foragers. pnas.org/doi/10.1073/pnas.1419966111

Mesolimbic SEEKING system and dopamine signaling in compulsive behavior. frontiersin.org/journals/human-neuroscience/articles/10.3389/fnhum.2021.635932/full

Recent review on the neuroscience of addiction, treatment, and prevention. pmc.ncbi.nlm.nih.gov/articles/PMC11600381/

Stanford research summary on addiction as a brain disorder and emerging treatment directions. news.stanford.edu/stories/2025/08/addiction-brains-science-treatment-prevention-research

End of framework draft.

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